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Hydrogen storage in lithium hydride: A theoretical approach

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ABSTRACT

First principles calculations have been carried out to analyze structural stability of lithium hydride (LiH) in NaCl phase using the full potential linearized augmented plane wave (FP-LAPW) method within the framework of density functional theory (DFT). Calculations have been extended to physiosorbed H-atom compounds $LiH{\cdot}H_2$, $LiH{\cdot}3H_2$ and $LiH{\cdot}4H_2$. The obtained results are discussed in the paper. The results for LiH are in excellent agreement with earlier reported data. The obtained direct energy band gap of LiH is 3.0 eV which is in excellent agreement with earlier reported theoretical band gap. The electronic band structure plots of the hydrogen adsorbed compounds show metallic behavior. The elastic constants, anisotropy factor, shear modulus, Young's modulus, Poisson's ratio and cohesive energies of all the compounds are calculated. Calculation of the optical spectra such as the real and imaginary parts of dielectric function, optical reflectivity, absorption coefficient, optical conductivity, refractive index, extinction coefficient and electron energy loss are performed for the energy range 0–15 eV. The obtained results for $LiH{\cdot}H_2$, $LiH{\cdot}3H_2$ and $LiH{\cdot}4H_2$, are reported for the first time. This study has been made in search of materials for hydrogen storage. It is concluded that LiH is a promising material for hydrogen storage.

1. Introduction

Hydrogen is considered as a possible substitute for fossil fuels because it is the most copious element in the universe; it is renewable and possesses high substance energy [1]. The large scale utilization of hydrogen as a fuel crucially depends on the development of compact storage materials with a high mass content of hydrogen [2]. The main problem in building a future hydrogen-based energy system is to find out safe methods to store large amount of hydrogen. The global energy demand has increased exponentially and is expected to increase by 75% in the period 2000–2030. In this regard Danish scientist, Poul La Cour (1846–1908) was the first person who gave an idea to create a hydrogen society [3].

The highest volumetric densities of hydrogen are found in metal hydrides. Many metals and alloys are capable of reversibly absorbing large amounts of hydrogen. The group one, two and three light metals (e.g. Li, Mg, B, Al) can combine with hydrogen to form a large variety of metal-hydrogen complexes. These are especially interesting because of their light weight and because of the number of hydrogen atoms per metal atom, which is two in many cases [4,5]. The metal hydrides for hydrogen storage need to be able to form hydrides with a high hydrogen-to-metal mass ratio, but they should not be too stable, so that the hydrogen can easily be released without excessive heating [6]. The ideal hydrogen storage material should have at least six major criteria: gravimetric capacity, volumetric capacity, kinetics, heat transfer, energy efficiency and reversibility.

On these grounds alkali metal hydride compound, e.g. LiH can be used for hydrogen storage [7]. Sigalas in his recent paper [8] has mentioned that LiH possesses the highest hydrogen content than any other hydride. LiH has also attracted much attention because of their use in nuclear industry [9]. LiH is the simplest alkali hydride, crystallizes in NaCl phase (Space Group No. 225, *Fm-3m*) and its electronic properties have been a subject of wide-range investigation [10–13].

For mobile and stationary applications the volumetric and gravimetric density of hydrogen in a storage material is crucial. Zuettel [4] has mentioned that there are six options for hydrogen storage: (1) high-pressure gas cylinders, (2) liquid hydrogen in cryogenic tanks, (3) adsorbed hydrogen on materials with a large specific surface area (at $T < 100$ K), (4) absorbed on interstitial sites in a host metal (at ambient pressure and temperature), (5) chemically bonded in covalent and ionic compounds (at ambient pressure), or (6) through oxidation of reactive metals, e.g. Li, Na, Mg, Al, Zn with water. Zurek et al. [14] have mentioned that all of the high LiH phases with unusual stoichiometries (LiH_n, $n = 2, 6, 8$) become enthalpically stable relative to LiH and H_2.

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There are remarkable structural varieties in the stable LiH_n; some contain only H_2 units, some both H_2 and H— (or a host LiH lattice with H_2 guests). They have shown that stability range of LiH_2, LiH_6 and LiH_8 are, respectively, 130–300+, 140–300+ and 100–200 GPa.

Recently Siqi Shi et al. [15] published a paper on Multi-scale computation methods: their applications in lithium-ion battery research and development. They have nicely mentioned that how simulation supplements to experimental efforts in order to bridge experimental and technological gaps. First-principles study of hydrogen storage on graphene with Li atoms by Weiwei Zhou et al. [16] shows that hydrogen storage capacity can be increased to 16 wt % by adjusting the coverage of Li atoms on graphene.

The above mentioned studies motivated us to study hydrogen storage in LiH based on first-principles calculations within the framework of density functional theory (DFT) using the full potential linearized augmented plane wave (FP-LAPW) method as DFT is the most powerful tools for carrying out theoretical studies with high accuracy [17,18]. Therefore, we carried out investigations related to the structural and electronic stabilities of LiH, LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ in space group no. 225, *Fm-3m* employing DFT and FP-LAPW method. Here our aim is that after adsorption of hydrogen the structure of host LiH will not be change. There is no evidence of previous work for LiH with extra added hydrogen residing in interstitial region at ambient condition. We believe that our results will serve as a good reference data for the application purposes.

2. Computational details

The calculations have been performed using FP-LAPW approach within the framework of DFT as implemented in the Wien2k package [19]. The Kohn-Sham equations are solved using a basis of linearized augmented plane wave [20]. The generalized gradient approximation (GGA) is adopted for the exchange-correlation energy [21]. The dependency of the total energy on the number of k-points in the irreducible wedge of Brillouin zone has been optimized and final size of the k-point mesh is set equal to 10*10*10. A maximum angular momentum for the radial wave function, l_{max}, has been taken as 10 and plane wave cutoff parameter, $R_{MT}{}^{*}K_{MAX}$ is set equal to 7 for all the compounds, where R_{MT} is the smallest radius of the muffin-tin spheres in unit cell and k_{max} is the magnitude of the largest k-vector in the reciprocal lattice. The iteration process was repeated until the calculated total energy and charge of the crystal converges to less than 0.0001Ry and $0.001e^{-}$, respectively. The bulk LiH exists in B1 type structure in ground state. The Wyckoff position of atom Li is 2a (0, 0, 0) and that of atom H is 2c (1/2, 1/2, 1/2) [22,23].

3. Results and discussion

3.1. Structural properties

In order to optimize unit cell volume and corresponding total energy of studied compounds, a set of volume versus energy has been obtained using Wien2k and fitted to Murnaghan's equation of state. The exchange and correlation functional is solved using the GGA due to Perdew and Wang [24]. The other equilibrium unit cell parameters are determined by these fits for LiH, LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$.

Muffin tin radii (R_{MT}) of these atoms play an important role in determining various physical properties [25]. The used values of R_{MT} of constituent atoms and their atomic positions are listed in Table 1. The obtained unit cell crystal structures of LiH, LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ are shown in Fig. 1(a)–(d), respectively. In figures Li atoms are shown in green color, basic hydrogen atoms in violet color and extra added hydrogen in red color.

Fig. 1(a) shows unit cell structure of LiH. This unit cell contains 8*(1/8) + 6*(1/2) = 4 atoms of Li and 12*(1/4) + 1 = 4 atoms of H. Thus one unit cell contains *four formula units* of LiH. Fig. 1(b) shows unit cell structure of LiH along with 8 added hydrogen atoms residing inside unit cell exactly in middle of Li-H layer. Distance between two nearest added hydrogen atoms is 2.18 Å. The number of added hydrogen atoms per formula is two (1 hydrogen molecule). Similarly Fig. 1(c) shows unit cell structure of LiH along with 12 added hydrogen atoms residing inside unit cell and 24*(1/2) = 12 atoms on surface of unit cell. That is, in this figure, effective number of added H-atoms is 24, meaning thereby six added hydrogen atoms (3 hydrogen molecules) per formula. In this case too, between two adjacent layers of LiH, the added hydrogen atoms form a plane and in one plane 12 hydrogen atoms are arranged in five rows (two on surfaces and three in side unit cell). Also, Fig. 1(d) shows unit cell structure of LiH along with 32 added hydrogen atoms residing, completely, inside unit cell. Between two adjacent layers of Li-H all the hydrogen atoms are arranged in two layers and each layer contains two rows. In this case number of added hydrogen atoms per formula is eight (4 hydrogen molecules).

Table 1
The used values of muffin-tin radius (R_{MT}) and atomic positions of different atoms for LiH, LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ in B1 phase.

Compounds	Atom	R_{MT} (a.u.)	Atomic positions
LiH·$0H_2$	Li	1.7	0, 0, 0
	H	1.3	0.5, 0.5, 0.5
	Added H	–	
LiH·H_2	Li	1.8	0, 0, 0
	H	1.3	0.5, 0.5, 0.5
	Added H	1.1	0.25, 0.25, 0.75
LiH·$3H_2$	Li	1.9	0, 0, 0
	H	1.3	0.5, 0.5, 0.5
	Added H	1.0	0.00, 0.25, 0.25
LiH·$4H_2$	Li	2.0	0, 0, 0
	H	1.3	0.5, 0.5, 0.5
	Added H	1.1	0.65, 0.35, 0.35

The obtained equilibrium volume, lattice constant, bulk modulus and first derivative of the bulk modulus in LiH·nH_2 are summarized in Table 2, along with the available reported data for LiH. From this table it is clear that for LiH our predicted lattice constant 'a' (4.009 Å) is in excellent agreement with the earlier reported theoretical (4.009 Å) [26] and experimental value (4.083 Å) [27]. The obtained bulk modulus 'B_0' (35.76 GPa) is also comparable with its earlier reported value (33.6 GPa) [28]. The results for LiH·nH_2 (n = 1,3,4) are reported for the first time.

3.2. Cohesive energy

The total energies of LiH (−16.2719Ry), LiH·H_2 (−18.3807Ry), LiH·$3H_2$ (−22.6082Ry) and LiH·$4H_2$ (−24.9082Ry) have been optimized. The energy of isolated Li and H atoms are $E_{tot}(Li) = -15.04$Ry [29] and $E_{tot}(H) = -0.97$Ry [30]. The cohesive energy of the compounds is defined as:

$$\text{Cohesive } energy = (\text{Energy of free atoms} - \text{Crystal energy})$$

$$E_{coh}^{LiH} = \left(E_{tot}^{Li} + E_{tot}^{H}\right) - E_{tot}^{LiH} \quad (1)$$

Similarly cohesive energy of the other compounds with 2*n added hydrogen atom are calculated using formula

$$E_{coh}^{LiH\cdot n*H2} = \left(E_{tot}^{LiH} + 2{*}n * E_{tot}^{H}\right) - E_{tot}^{LiH\cdot n*H_2} \quad (2)$$

Thus, by using equation (1) the obtained cohesive energy for LiH (0.2610Ry) is in good agreement with theoretical predicted value, 0.396Ry of N Novakovic [31]. The obtained cohesive energies of other compounds are listed in Table 3.

From the table it is evident that total energy decreases with increase in total number of added hydrogen atoms. The amount of energy released per H-atom during physiosorption of two, six and eight H-atoms are respectively, 0.0844 Ry, 0.0860Ry and 0.1095 Ry. It is also evident from the table that cohesive energy of LiH (0.261Ry) is more than the energy released during addition of any single hydrogen atom. Thus to retrieve

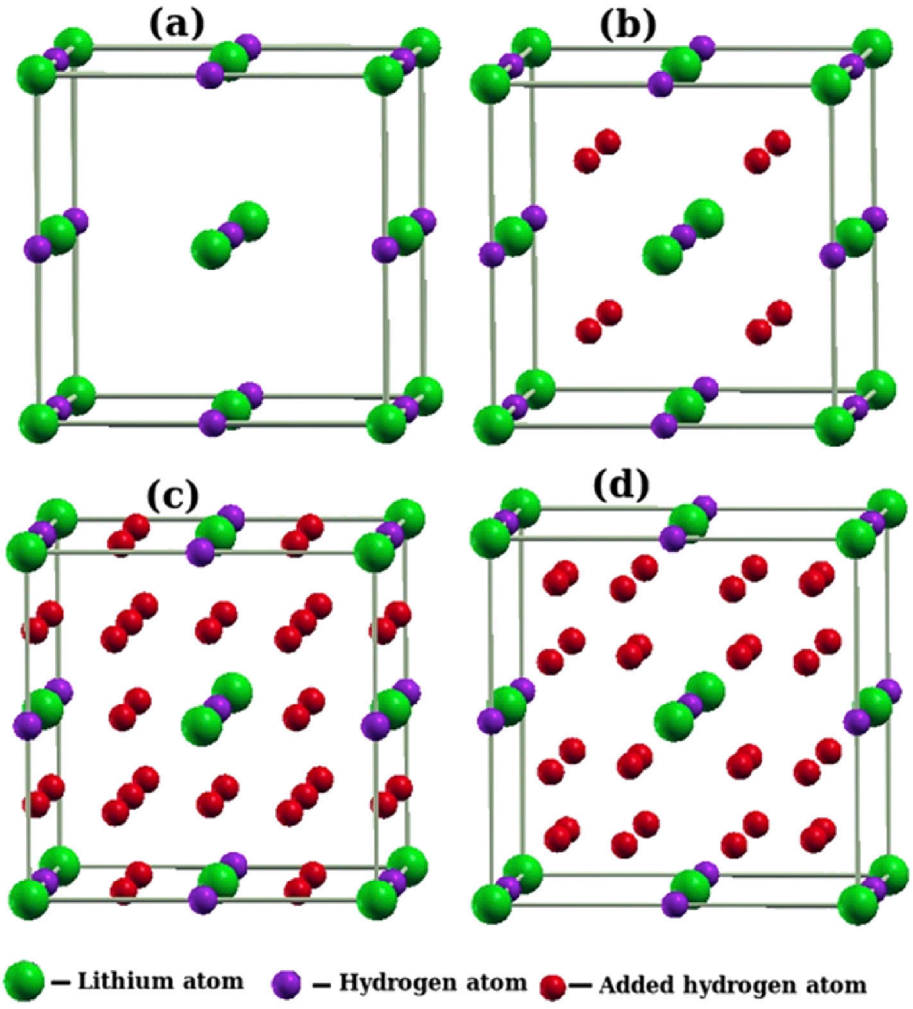


Fig. 1. Unit cell crystal structures of (a) LiH, (b) LiH·H_2, (c) LiH·$3H_2$ and (d) LiH·$4H_2$ in B1 structure.

Table 2
Optimized equilibrium cell volume, lattice parameter, bulk modulus and first pressure derivative of bulk modulus for LiH·nH_2 (n = 0,1,3,4) in B1 phase.

Material	Vo (in Å^3)	'a' (in Å)	Bo (in GPa)	Bo′
LiH	16.10	4.007 4.083 [26] 4.008 [27]	35.76 33.60 [28]	3.1
LiH·H_2	20.53	4.348	56.23	1.3
LiH·$3H_2$	32.89	5.090	72.29	3.2
LiH·$4H_2$	39.60	5.410	75.61	3.3

hydrogen atoms when external energy will be supplied to the system added hydrogen atoms will be released first.

3.3. Electronic properties

In order to study electronic properties of LiH·nH_2 (n = 0,1,3,4), their band structure and density of states (DOS) plots have been studied. The band structure plot for LiH calculated using PBE-GGA is shown in Fig. 2(a). The obtained direct energy band gap corresponding to X-symmetry line is 3.0 eV. Our reported direct energy band gap is in excellent agreement with the earlier reported theoretical band gap ≈3.0 eV [32] but less than experimentally obtained energy band gap of 4.9 eV [33]. The band structures plotted in Fig. 2(b)–(d) show metallic

Table 3
Optimized total energy, energy released during addition of H atoms and cohesive energy (E_{coh}) of LiH·nH_2 (n = 0,1,3,4) in B1 structure along with total energy of atom Li and H.

Materials	Energy (Ry)	Added H-atoms	Energy (Ry) Li + H	Energy (Ry) LiH·nH	Cohesive Energy (Ry)		Energy released during addition of H atom (Ry)
					LiH	LiH·nH	
Li	−15.04 [29]						–
H	−0.97 [30]						–
LiH	−16.2719		−16.01		0.2610		–
LiH·H_2	−18.3807	2		−18.2119		0.1688	0.0844
LiH·$3H_2$	−22.6082	6		−22.0919		0.5163	0.0860
LiH·$4H_2$	−24.9082	8		−24.0319		0.8763	0.1095

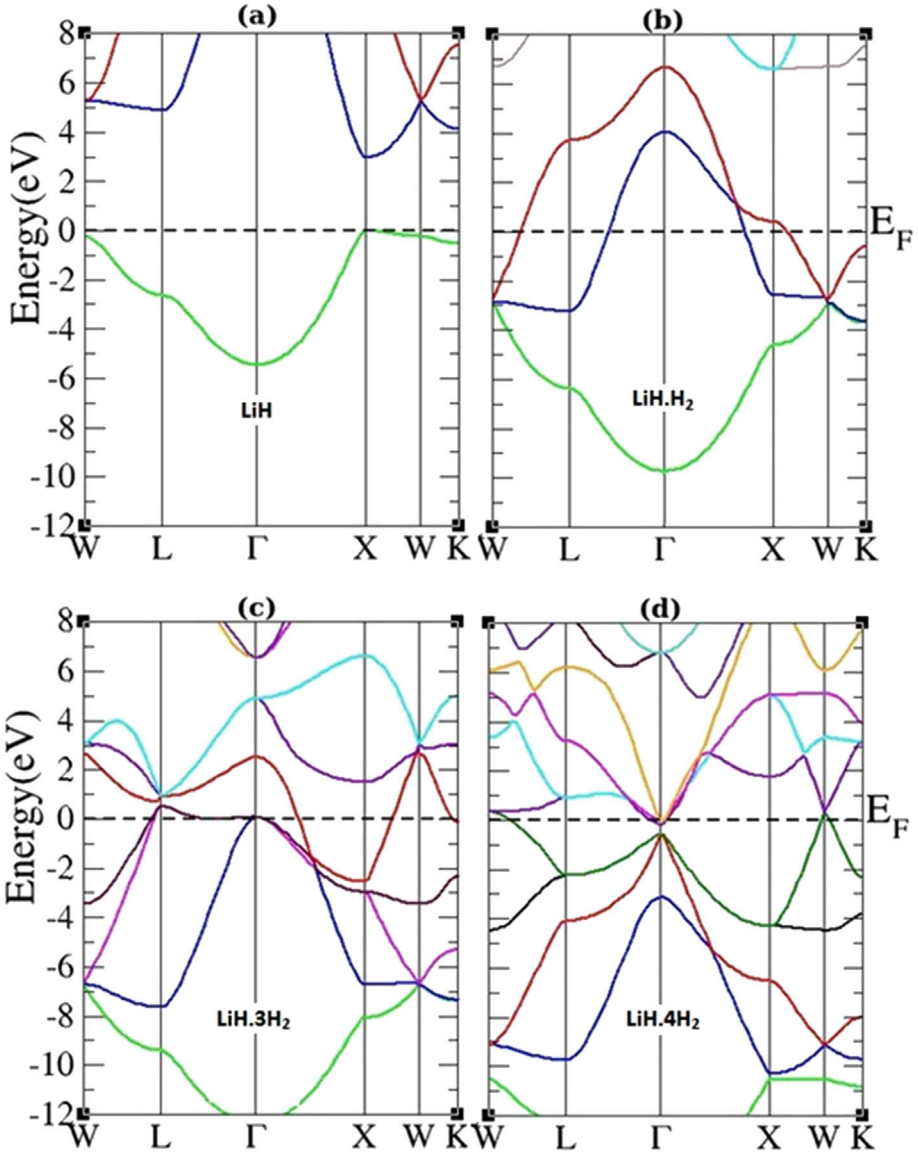


Fig. 2. Electronic band structures plots for (a) LiH, (b) LiH·H_2, (c) LiH·$3H_2$ and (d) LiH·$4H_2$.

characters of LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$, respectively. From the band structures of added hydrogen compounds, it is evident that both the valance bands as well as conduction bands shifted downward.

In order to understand nature of bonding and charge distribution after addition of extra hydrogen atoms in LiH we have also plotted band structures showing *s*-character bands of Li, H and added H along with their *total*-character bands in Figs. (3–5) for LiH·nH_2 (n = 1,3,4). From the plots, it is very clear that both the *s*-character as well as the *total*-character bands of Li and H are almost identical. But the *s*-character as well as the *total*-character bands of the added H are quite different. In Figs. (3–5) the size of circles on the bands indicate the population of charge, which is directly proportional to the size of circles. Distribution of charge related to the added hydrogen atoms is not confined to *s*-orbitals because total character band charge density is highly populated in comparison to s-orbital charge. It seems that electrons in the added hydrogen atoms are not confined to only s-orbitals but they are excited to the higher orbitals. In this way an induced dipole may be developed on hydrogen atoms that makes them different and Van der Wall type forces binds them together. Electron charge density plots for LiH, LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$, shown in Fig. 6, supports above conclusion as in this figure volume of electron charge around added hydrogen is more than that involved in Li-H bonding.

Further in Fig. 6 equal charge density surfaces are drawn for all the studied compounds with a contour level of 0.1 e/Å^3. It is noticeable from Fig. 6(a) that LiH has ionic bonding, as in this case the charge density distribution is spherical around the Li— and H-atoms. It implies that the valence electron of Li is transferred to the hydrogen atom. But, in case of LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ (Fig. 6(b–d)), charge density around Li and H atoms are not symmetric which shows considerable sharing among Li—, H-atoms. It clearly indicates covalent bonding between Li and H atoms.

The total and partial DOS plots for all the cases are shown in Fig. 7. All the findings included in above paragraph are verified from their DOS plots. In case of LiH below Fermi energy H-atoms dominates in DOS, while in case of added H compounds the domination of added H is prominent upto −4 eV.

3.4. Elastic properties

The elastic properties provide information about mechanical and

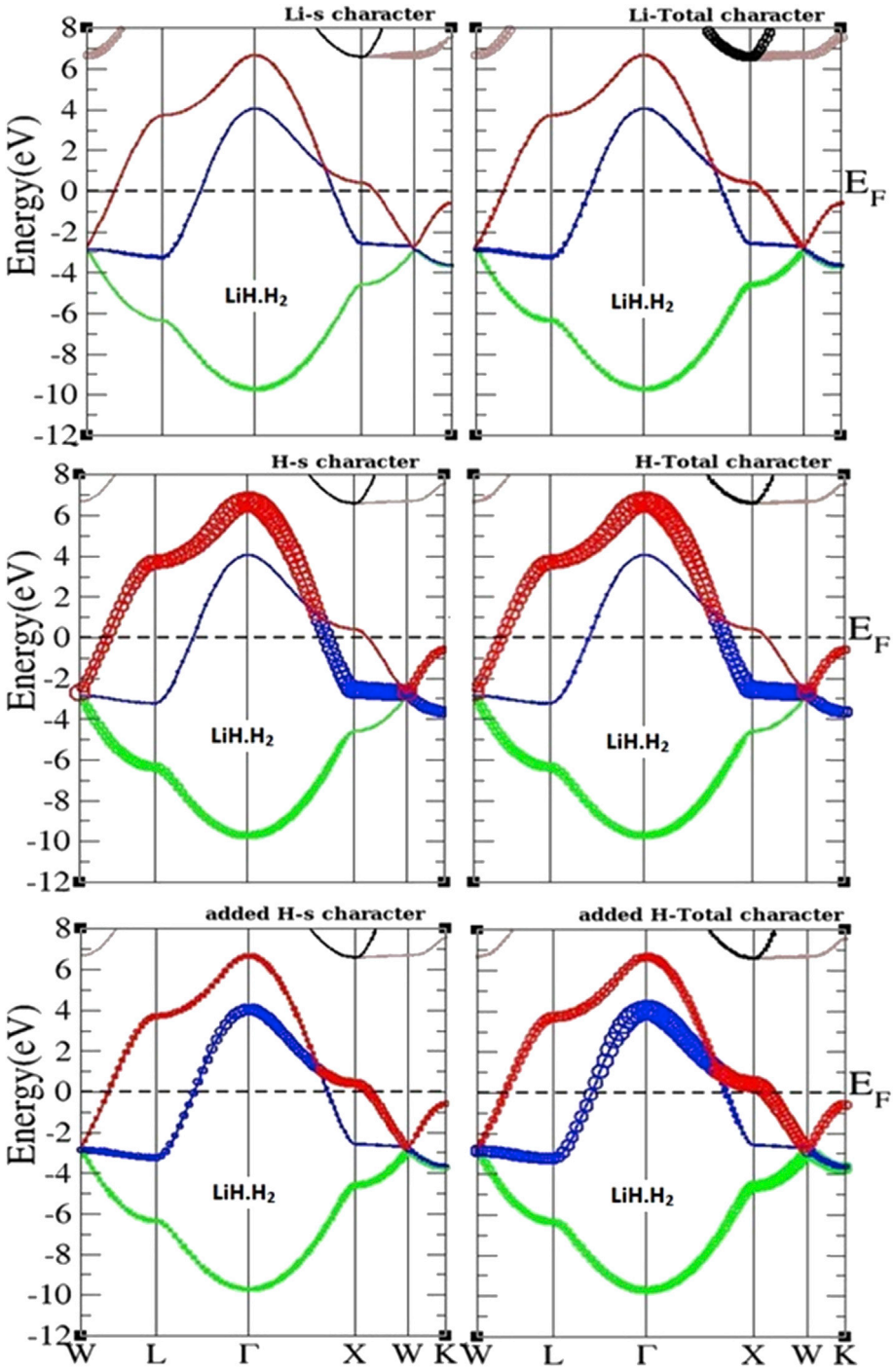


Fig. 3. The *s*-character as well as the *total*-character bands of Li, H and added H in LiH·H_2.

dynamical behaviour of crystals; the forces operating in solids provide important data for developing inter-atomic potentials and it define how a material undergoes stress deformations and then recovers and return to its original shape after stress ceases [34]. The elastic constants are important parameters of a material and give information about the structural stability, the bonding character between adjacent atomic planes, and anisotropic character. The cubic crystal has three different elastic coefficients (C_{11}, C_{12}, and C_{44}). The elastic constants are evaluated using Born's criteria [35]. The elastic constants of LiH·nH_2 (n = 0,1,3,4) at their equilibrium volumes are determined from the First-principles calculations. The calculated elastic constants are positive and satisfy the mechanical stability criteria $C_{11} > 0$, $C_{11} - C_{12} > 0$, $C_{44} > 0$, $C_{11} + 2C_{12} > 0$ [36] and the bulk modulus B also fulfilling the criterion: $C_{12} < B < C_{11}$.

Once three elastic constants namely C_{11}, C_{12}, and C_{44} have been calculated; bulk modulus, B; Young's modulus, Y; isotropic shear modulus, G; Poisson ratio, υ (which are important fundamental parameter closely connected to many physical properties such as internal strain, thermoelastic stress, sound velocity, fracture toughness) can be calculated using the expressions given below [37]:

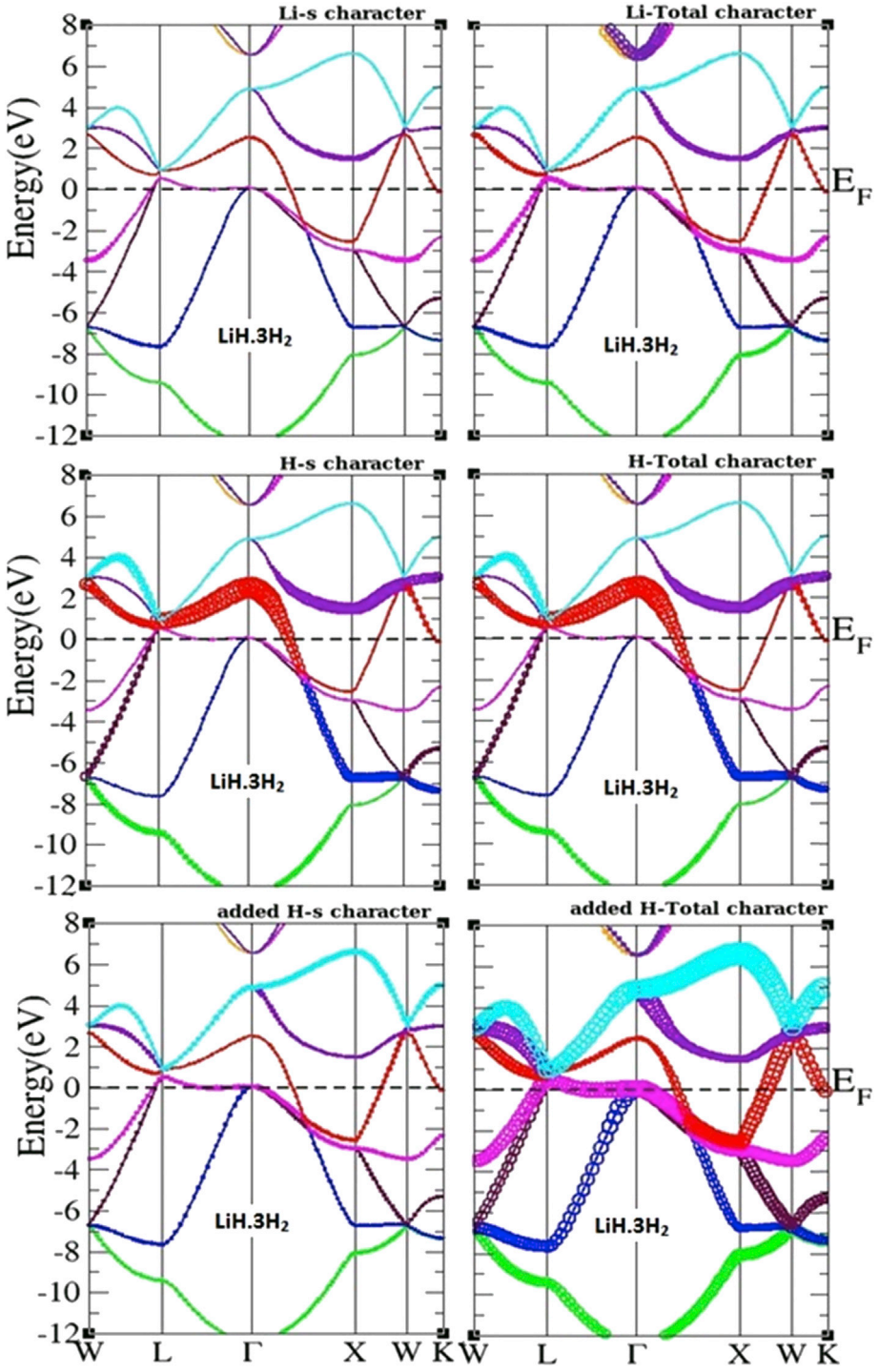


Fig. 4. The *s*-character as well as the *total*-character bands of Li, H and added H in $LiH{\cdot}3H_2$.

$$G = \left(\frac{G_V + G_R}{2}\right) \tag{3}$$

Here, G_V is Voigt's shear modulus corresponding to the upper bound of G values, and G_R is Reuss's shear modulus for cubic crystals corresponding to the lower bound values, expressed as:

$$G = \left(\frac{G_V + G_R}{2}\right) \tag{4}$$

$$G_R = \frac{5[(C_{11} - C_{12})4C_{44}]}{4C_{44} + 3(C_{11} - C_{12})} \tag{5}$$

Young's modulus (Y) is a good indicator about the stiffness of the material. When it is higher for a given material, the material is stiffer.

The expression for the Young's modulus is given by:

$$Y = \frac{9GB}{G + 3B}\frac{9GB}{G + 3G} \tag{6}$$

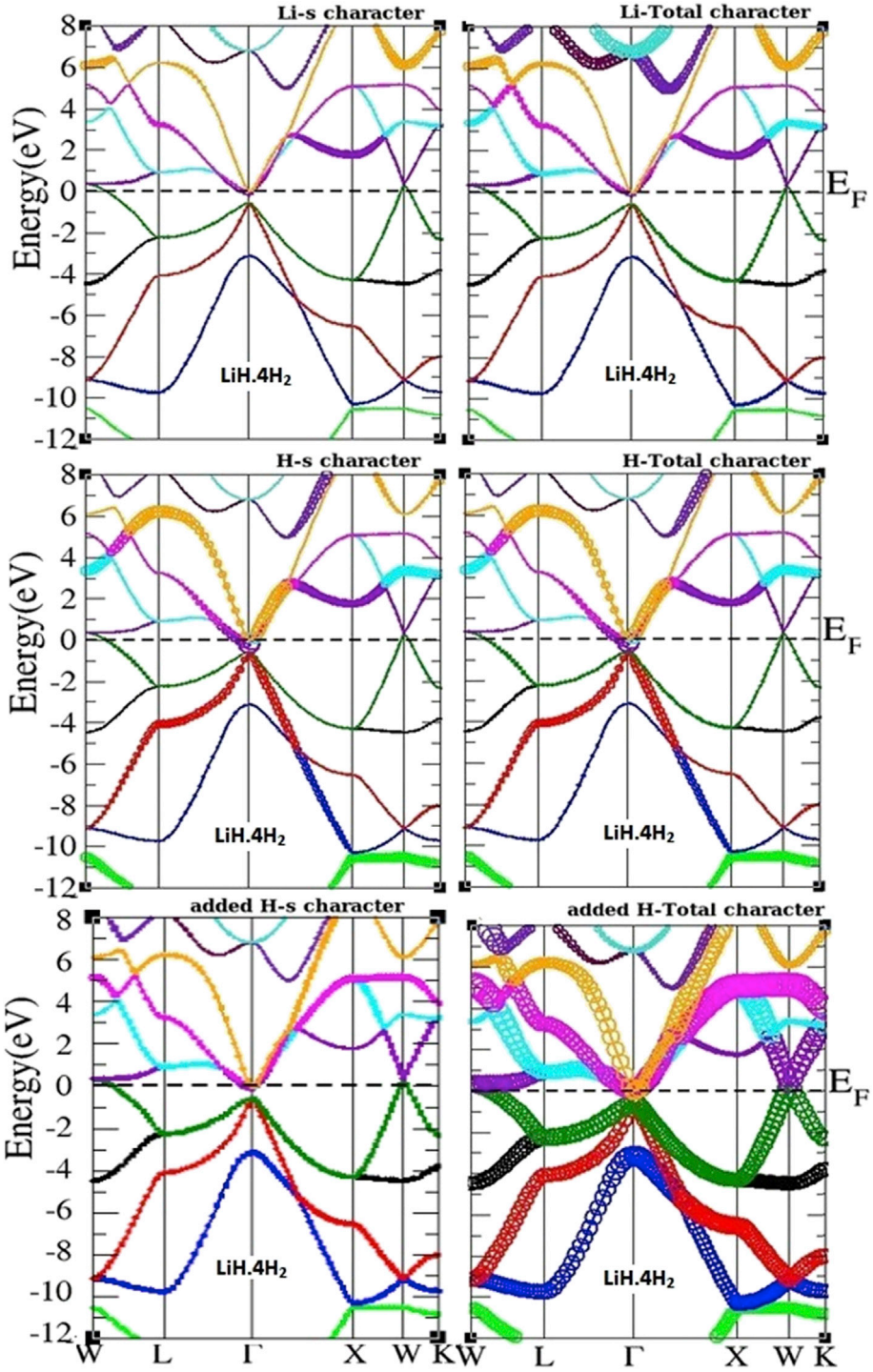


Fig. 5. The *s*-character as well as the *total*-character bands of Li, H and added H in LiH·$4H_2$.

Poisson's ratio provides more information for dealing with the characteristic of the bonding forces. The value of the Poisson ratio (υ) for covalent materials is small (υ < 0.1), whereas for ionic materials a typical value is 0.25 [38].

The expression for the Poisson's ratio is given by:

$$\upsilon = \left(\frac{3B - Y}{6B}\right) = \frac{C_{12}}{C_{11} + C_{12}} \quad (7)$$

In our calculations υ are 0.18, 0.16 and 0.31 in cases of LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$, respectively. Hence, a higher ionic contribution in an inter-atomic bonding for these compounds increases with the increase in number of added hydrogen atoms. In Fig. 7 electronic charge distribution around Li and H are perfectly spherical while in case of added hydrogen compounds slid deformation in charge distribution is seen but charge distribution around added hydrogen is spherical.

The elastic anisotropy of a cubic crystal can be characterized by the

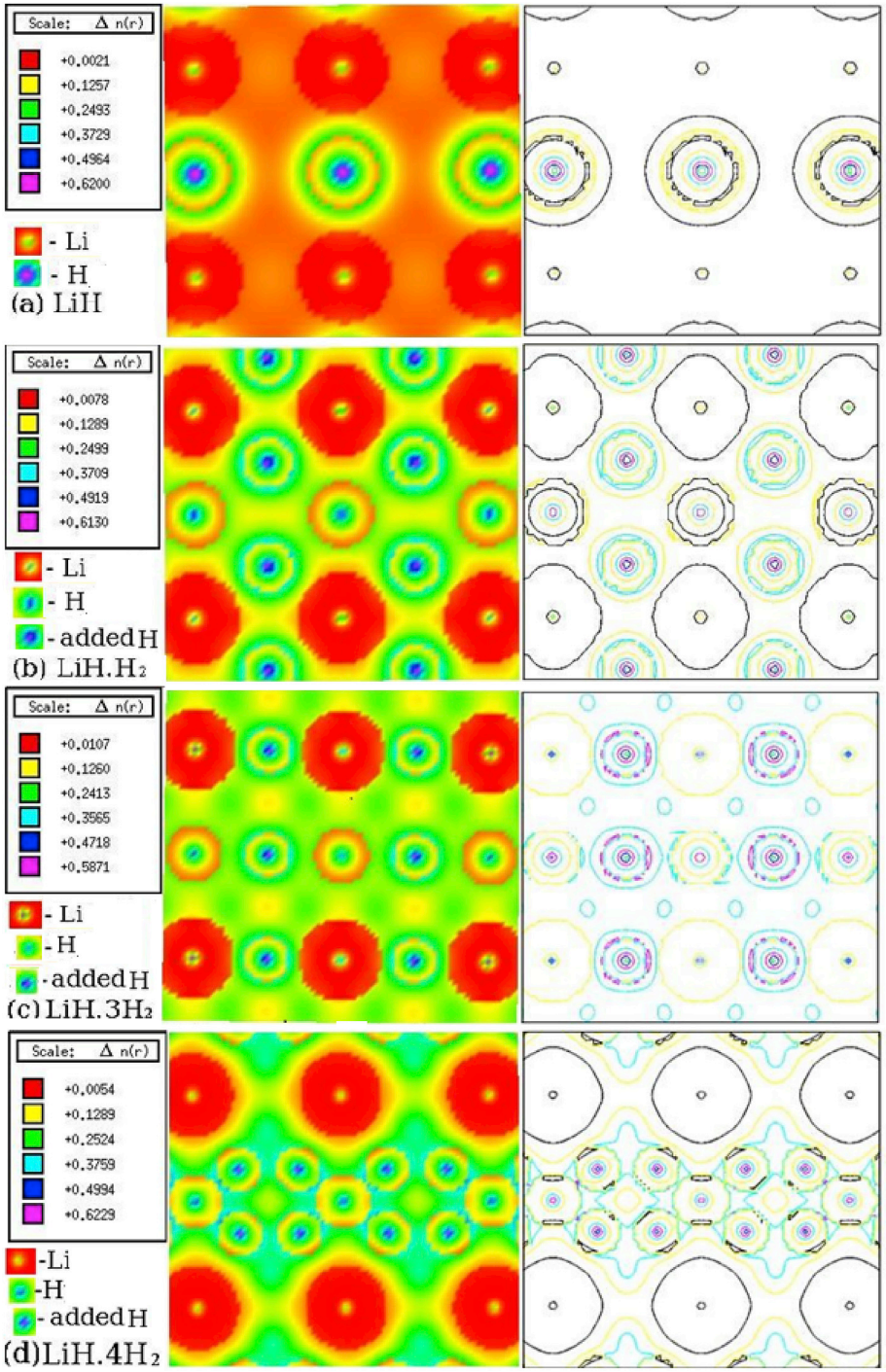


Fig. 6. Electron charge density plots in (111) plane for (a) LiH, (b) LiH·H_2, (c) LiH·3H_2, (d) LiH·4H_2.

Zener anisotropy factor, *A*, which represents the ratio of the two elastic-shear coefficients:

$$A = 2\frac{C_{44}}{(C_{11} - C_{12})} \tag{8}$$

The anisotropy factor (*A*) is equal to one for an isotropic material, while any value smaller or larger than one indicates anisotropy. The obtained values of the anisotropy factor *A* are 3.49, 0.47, 1.00 and 1.88 in case of LiH, LiH·H_2, LiH·3H_2 and LiH·4H_2, respectively. Thus the behaviour of LiH·3H_2 seems to be isotropic and rest of the compounds are anisotropic.

The bulk modulus calculated from the elastic constants $B = \frac{1}{3}(C_{11} + 2C_{12})$ is in good agreement with that obtained from the total energy minimization calculations (see Table 4). Mechanical properties such as ductility and brittleness of materials can be explained from the proposed relationship. The Shear modulus G represents the resistance to plastic deformation, while the bulk modulus B represents the resistance to fracture. We know that there is a criterion for B/G ratio ($R_{B/G}$) which separates the ductility and brittleness of materials. According to Pugh's criteria [39], if B/G > 1.75 the material is ductile, otherwise it is brittle.

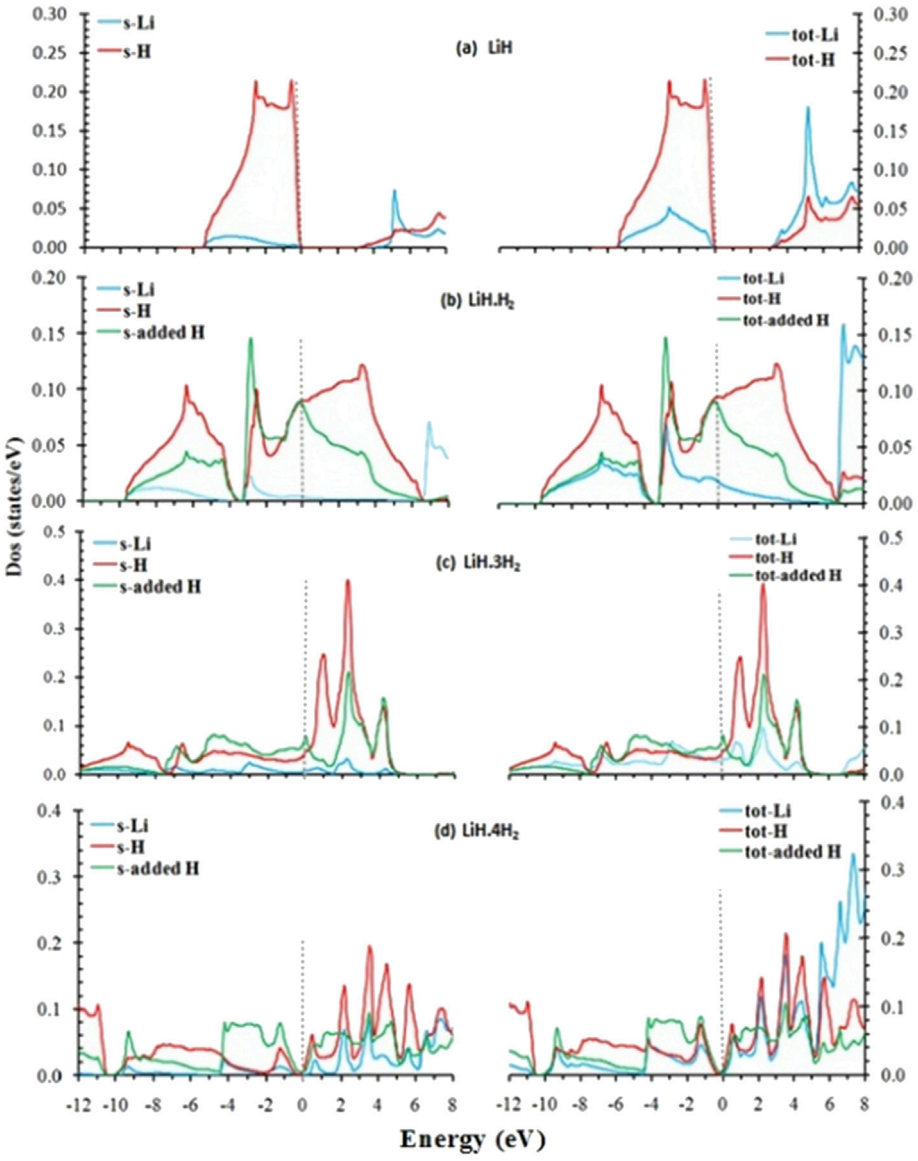


Fig. 7. Plots of the *s*-state and total DOS of atoms Li, H and added H in LiH·nH_2 (n = 0,1,3,4). Fermi levels are set at 0 eV.

LiH, LiH·H_2 and LiH·3H_2 indicate brittle behavior due to the fact that values of $R_{B/G}$ are 0.46 and 1.23 and 1.15, respectively, but LiH·4H_2 behaves in a ductile manner as its $R_{B/G}$ is 2.32.

From Table 4, it is analyzed that calculated elastic constants of LiH are very close to the experimental [40,41] and other theoretical values [42–44]. There is no experimental or theoretical evidences are found for the elastic constants of the compounds LiH·nH_2 (n = 1,3,4); hence our results can serve as a reference for future investigations.

3.5. Optical properties

The optical properties for the studied compounds have been calculated using OPTIC package of Wien2k code [45]. In general the optical properties of matter can be described by means of the complex dielectric functions $\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega)$, where $\varepsilon_1(\omega)$ and $\varepsilon_2(\omega)$ describes the dispersive (real part) and absorptive parts (imaginary part) of the dielectric function. For cubic symmetry, we only need to calculate one

Table 4
Calculated elastic constants C_{11}, C_{12}, and C_{44} (in GPa), Shear constant (G), Zener anisotropy factor (*A*), Young's modulus (Y), Poisson ratio (*v*) and ratio of bulk modulus to shear constant in B1 structure.

Material/ Compounds	C_{11}	C_{12}	C_{44}	G	'A'	'Y'	*v*	$R_{B/G}$ ratio
LiH	79.628	12.831	50.654	78.01	3.49	136.15	0.13	0.46
	77.0 [42]	12.7 [42]	48.2 [42]					
LiH·H_2	92.40	35.79	13.41	44.17	0.47	104.39	0.18	1.27
LiH·3H_2	96.26	49.85	24.23	54.96	1.0	128.07	0.16	1.31
LiH·4H_2	102.11	62.89	37.06	32.62	1.88	85.60	0.31	2.32

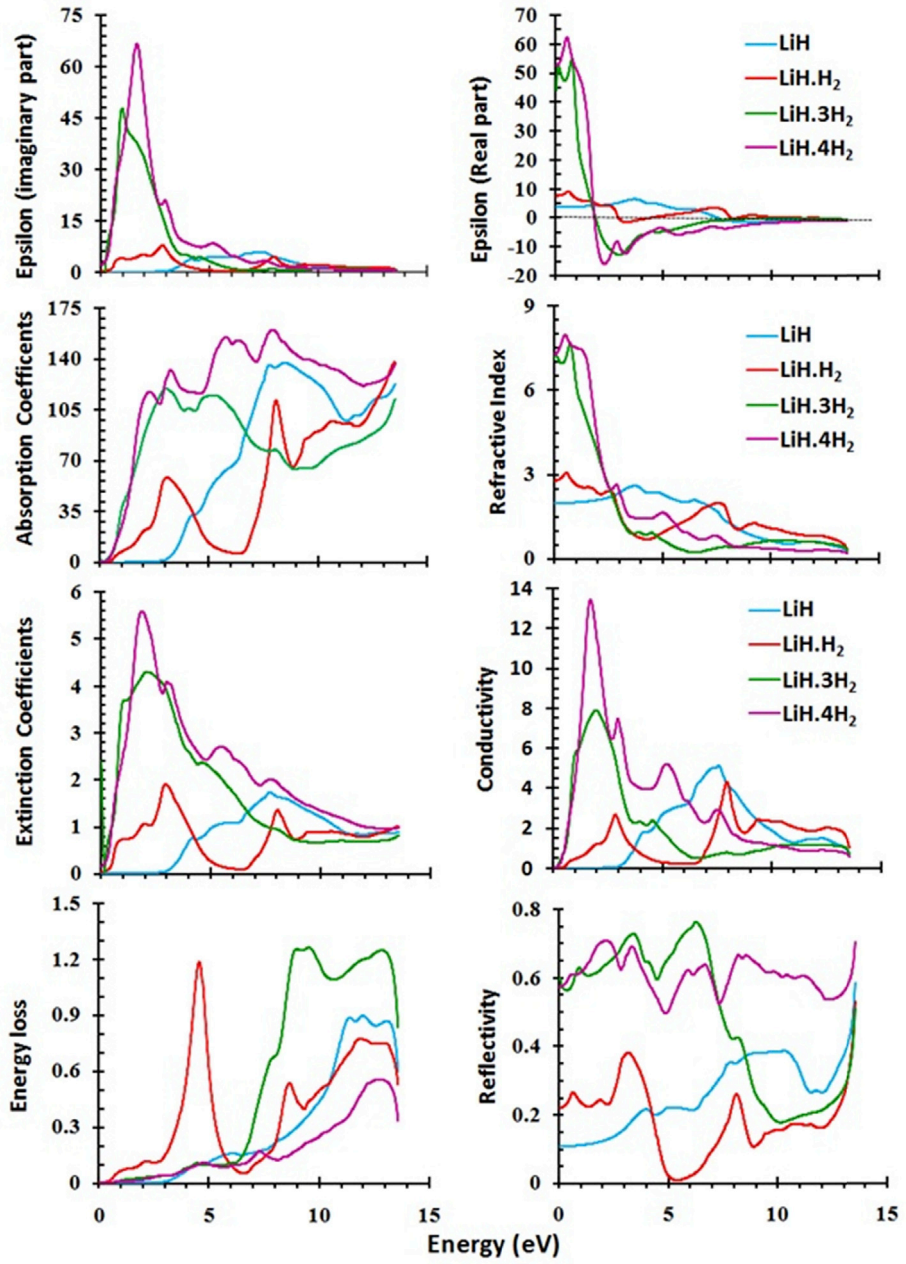


Fig. 8. Optical parameters of LiH·nH$_2$ (n = 0,1,3,4), plotted as functions of energy: (a) imaginary parts of the dielectric constants, (b) real parts of the dielectric constants, (c) absorption coefficients, (d) refractive indices, (e) extinction coefficients, (f) conductivity, (g) energy loss functions and (h) reflectivity.

dielectric tensor component to completely characterize the linear optical properties. The imaginary part $\varepsilon_2(\omega)$ of the frequency-dependent dielectric function is given by Ref. [46].

$$\varepsilon_2(\omega) = \frac{Ve^2}{2\pi\hbar\omega^2 m^2}\int\sum|\psi_C|p|\psi_V|\delta\,(E_C - E_V - \hbar\omega)d^3\kappa \tag{9}$$

here ψ_C and ψ_V are the wave functions in the conduction and valance bands, p is the momentum operator, ω is the photon frequency, $\hbar$ is reduced Planck's constant, V is the volume of the crystal, e is the electron charge. The real part $\varepsilon_1(\omega)$ of the dielectric function can be evaluated from $\varepsilon_2(\omega)$ by using the Kramer-Kroning relations [47,48].

$$\varepsilon_1(\omega) = 1 + \frac{2}{\pi}P\int_0^{\infty}\frac{\omega'\varepsilon_2(\omega')}{\omega'^2 - \omega^2} \tag{10}$$

where 'P' is the principle value of the integral. The knowledge of both the real and imaginary parts of the dielectric function allows the calculation of the important optical properties such as refractive index n (ω) through the following equation:

$$n(\omega) = \frac{1}{\sqrt{2}}\left[\sqrt{\varepsilon_1(\omega)^2 + \varepsilon_2(\omega)^2} + \varepsilon_1(\omega)\right]^{1/2} \tag{11}$$

The expression for all other frequency dependent optical properties, such as the extinction coefficient $k(\omega)$, the absorption coefficient $\alpha(\omega)$, the optical reflectivity R(ω), the energy loss spectrum L(ω) and the optical conductivity $\sigma(\omega)$ are obtained through the dielectric function $\varepsilon(\omega)$. Required equations for above mentioned parameters are as follows:

Table 5
The values of dielectric constant, refractive index and reflectivity (%).

Optical Parameters	Compounds			
	LiH	LiH·H_2	LiH·$3H_2$	LiH·$4H_2$
Dielectric constant	3.90	7.94	44.40	52.54
Refractive index	1.97	2.77	7.20	7.26
Reflectivity (%)	10.75	22.11	59.77	57.41

$$\text{Extinction coefficient } k(\omega) = \frac{\varepsilon_2(\omega)}{2n(\omega)} \tag{12}$$

$$\text{Absorption coefficient } \alpha(\omega) = \sqrt{2\omega}\left[\sqrt{\varepsilon_1^2(\omega)+\varepsilon_2^2(\omega)} - \varepsilon_1(\omega)\right]^{1/2} \tag{13}$$

$$\text{Optical reflectivity } R(\omega) = \left|\frac{1-\sqrt{\varepsilon(\omega)}}{1+\sqrt{\varepsilon(\omega)}}\right|^2 \tag{14}$$

$$\text{Energy loss spectrum } L(\omega) = \frac{\varepsilon_2(\omega)}{[\varepsilon_1^2(\omega)+\varepsilon_2^2(\omega)]} \tag{15}$$

$$\text{Optical conductivity } \sigma(\omega) = \frac{\omega}{4\pi}\varepsilon_2(\omega) \tag{16}$$

Our calculated optical properties at the equilibrium lattice constant for LiH·nH_2 (n = 0,1,3,4) are presented for the energy range 0–15 eV. The plotted graphs are shown in Fig. 8.

From Fig. 8(a) the analysis of imaginary part $\varepsilon_2(\omega)$ shows that the threshold energy of the dielectric function occurs at 3.0 eV for LiH which corresponds to the optical transitions from the highest valence band to the lowest conduction band. The prominent peaks of magnitude (in arbitrary units) 5.60, 6.99, 46.97 and 66.17 are located at 7.41 eV, 2.95 eV, 1.02 eV and 1.67 eV for LiH·nH_2 (n = 0,1,3,4), respectively. The real part of $\varepsilon_1(\omega)$ is shown in Fig. 8(b). In the plots the magnitude of main peaks are: 6.311 (LiH), 8.24 (LiH·H_2), 53.90 (LiH·$3H_2$), and 61.77 (LiH·$4H_2$) occur at 3.87 eV, 0.72 eV, 0.80 eV and 0.55 eV. The dielectric constant of a material is frequency dependent, to which both the electrons and the ions contribute. Here, we have only calculated the electronic contribution to the dielectric constant; this dielectric constant is termed as high-frequency dielectric constant. The values of static dielectric constants are obtained from zero frequency limit of the real part. The calculated values of the dielectric constant are 3.9, 7.94, 44.4 and 52.54, respectively for LiH·nH_2 (n = 0,1,3,4).

From Fig. 8(c), it is clear that the values of the optical absorption edges are located at 3.0 eV for LiH and 0 eV for others due to their metallic nature. The optical absorption edges located at 3.0 eV for LiH is due to interband transition. The value of optical absorption of LiH is in agreement to the available result [49].

The refractive index and the extinction coefficient are displayed in Fig. 8(d) and (e). The compounds with high refractive index have high optical non-linearity. The obtained values of refractive indices are 1.97, 2.77, 7.20 and 7.26 for LiH·nH_2 (n = 0,1,3,4), respectively. This shows that LiH·$3H_2$ possesses higher optical non-linearity as compared to others. The refractive index increases with increase in energy and reaches a peak of magnitude 2.588, 3.045, 7.612 and 7.930 (arbitrary unit) in visible region at about 3.8 eV, 0.63 eV, 0.80 eV and 0.58 eV for LiH·nH_2 (n = 0,1,3,4), respectively. The values of extinction coefficients 1.73, 1.9, 2.66 and 5.6 are obtained at 7.7 eV, 2.95 eV, 2.1 eV and 1.9 eV, respectively, for LiH·nH_2 (n = 0,1,3,4).

In Fig. 8(f), optical conductivity spectra are plotted. These plots exhibit high conductivity in the energy range 1.5–8.0 eV with peaks at 7.74 eV, 3.01 eV, 2.14 eV and 1.92 eV, respectively, for LiH·nH_2 (n = 0,1,3,4). The optical conductivity is mainly caused by transitions from the H-*s* state of valence band to the conduction bands.

Electron energy loss function $L(\omega)$ describes the energy loss of a fast electron traversing in a material. In Fig. 8(g) peaks in $L(\omega)$ spectra represent the characteristic associated with the plasma resonance and the corresponding frequency is the so-called plasma frequencies, above which materials exhibit dielectric behaviour [$\varepsilon_1(\omega) > 0$] while below which material behaves the metallic property [$\varepsilon_1(\omega) < 0$]. That means, the positions of peaks in $L(\omega)$ spectra indicate also the point of transition from metallic to dielectric property for a material. At this energy, there is an abrupt reduction of the reflectivity, $R(\omega)$, spectrum, as shown in Fig. 8(h).

The numerical values of dielectric constant, refractive index and reflectivity (%) are listed in Table 5. From the table it is clear that all three parameters increases with the increase in the number of added hydrogens.

4. Conclusions

In this paper the structural, electronic, elastic and optical properties of LiH and LiH with added hydrogen atoms (LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$) have been investigated using density functional theory employing FP-LAPW approach together with the PEB-GGA in B1 structure. The equilibrium unit cell volume, lattice constant, bulk modulus, and pressure derivative of bulk modulus for above materials are reported. The electronic band structure and DOS are obtained and discussed at ambient condition. The obtained results for LiH are in excellent agreement with reported data. The obtained total energy decreases with increase in total number of added hydrogen atoms, implies that as we goes on increasing the added number of hydrogen the stability of compounds increases. The direct band gap in LiH is ~3.0 eV, while LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ are metallic in nature. The frequency dependent dielectric functions are calculated and analyzed. The optical properties confirm the insulating nature of LiH and metallic nature of LiH·H_2, LiH·$3H_2$ and LiH·$4H_2$ compounds. Thus, first-principle calculations offer an excellent way to predict new compounds for hydrogen storage or to obtain information on metal hydrides that is difficult to access experimentally.

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